Data-Driven Prediction of AI-Training Plateau Transitions

Morphology, recognition gates, remaining progress and selective forecasting

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Abstract

This paper develops a mathematical procedure for forecasting entry into a declared plateau morphology. It separates already-plateaued states from future transitions, endpoint morphology from first arrival, and slow progress from successful optimization. Causal target-specific landscapes, recognition gates and selective risk provide an application framework. Exact contraction, noisy-increment and counterexample trails show why plateau probability cannot be inferred from loss slope alone. The predictor is fitted to observed histories and matured labels, with target-specific comparisons to telemetry.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Causal observation and resolution contract

Let $\mathcal{F}_t$ contain the observations actually reported by issuance t: loss measurements, gradient and update statistics, known schedules and verified context. A delayed validation measurement belongs to the information set only when reported. The forecast horizon P is fixed by the query; W selects past extent, and m is a declared temporal sampling or internal-resolution parameter. These variables need not be functions of one another. For each target T, issue a forecast $p_{s,P,r}$

using only $\mathcal{F}_s$, with lens $r = (W, m)$. Once the label has matured, its loss can enter a causal surface

$$\widehat{L}_t(r; T, P) = \frac{\sum_{s \in \mathcal{A}_t} w_s \ell_T(Y^T_{s,P}, p_{s,P,r})}{\sum_{s \in \mathcal{A}_t} w_s}. \tag{1.1}$$

The same matured set $\mathcal{A}_t$ is used in numerator and denominator. Extend maturity when persistence or reporting delay requires follow-up beyond $s + P$. Unresolved negative outcomes are not known negatives. Empty history is missing risk, not zero risk. A pocket is a connected low-risk region under a specified adjacency and threshold. Retain each pocket's flow cross-sections, position, existence, signed errors and identity history. Fixed uses an eligible reference lens; Multi aggregates same-target alternatives; Auto changes a specified policy with causal context and matured utility. A pocket minimum is not an automatic event probability. An interpretable candidate event model is

$$p_{t,P} = \operatorname{logit}^{-1}\left(\beta_0 + \beta^\top z_t + \sum_i \alpha_i \phi(\mathcal{P}_i) + \sum_{i<j} \gamma_{ij} \psi(\mathcal{P}_i, \mathcal{P}_j)\right). \tag{1.2}$$

The coefficients, feature maps and any calibration map must be fitted from permitted data. Pocket labels must not have an artificial ordinal meaning; use a permutation-invariant aggregation or preserve a well-defined relational graph. Shared observations create dependence among pockets and cannot be counted as extra independent measurements. The compact archived AI representation used observable location/scale proxies in its multi-system scenario comparison, not explicit fits of all n/s parameters. A full scaling-spectrum branch is an optional extension that must calculate and qualify its own descriptors. Structural location/scale features and an actually fitted n/s spectrum are not interchangeable descriptions.

2 A plateau is a target definition, not a value judgment

A training plateau can mean low recent improvement, low gradient norm, small loss variability or small expected benefit per compute. These definitions are not equivalent. A high-loss constant trajectory and a low-loss constant trajectory can both have zero slope and variance. A morphology label alone does not certify convergence to a good solution or predict generalization.

Let b_t be an estimated slope and s_t an estimated variability over a fixed reported-loss window. Define current plateau morphology

$$A_t = \mathbf{1}\{|b_t| \le \varepsilon_b,\ s_t \le \varepsilon_s\}. \tag{2.1}$$

Thresholds, units, window size, fitting method and missing-data policy are part of the definition. A quantile boundary from development data is an operational convention, not a universal constant across architectures or tasks.

3 Future morphology and a genuine transition query

A future-endpoint transition label can be $Y^{\text{plat}}_{t,P} = A_{t+P}$ evaluated only when $A_t = 0$. This asks whether a nonplateaued current state will have plateau morphology at the future endpoint. If one instead defines $(1 - A_t)A_{t+P}$ on all states, already-plateaued cases are deterministic negatives for that algebraic label, not the same evaluation population as the restricted query.

For first arrival, define $\tau = \inf\{h \ge 1 : A_{t+h} = 1\}$ while initially at risk. Then the target is $\mathbb{P}(\tau \le P \mid \mathcal{F}_t, A_t = 0)$. A trajectory can enter and leave a plateau before $t + P$, so endpoint morphology and arrival risk need not agree.

Worked example 3.1 (Current recognition versus future entry)**.** Suppose $A_t = 1$. A model that always predicts current morphology can look accurate on an all-state future-plateau task, but it provides no evidence about entry from $A_t = 0$. Excluding current plateaus makes the forecasting question explicit. It does not by itself establish a general recognition gate for every unknown trajectory family.

4 Maturity and persistence indexing

If b_{t+P} and s_{t+P} use only a window ending at $t + P$, the label can mature then, subject to reporting delay. If plateau onset by P must be followed by d confirming increments, a late onset needs observations through $t + P + d$. The negative label is not mature at $t + P$ under that definition.

An alternative is to require the entire confirmation block to fit within the horizon; then eligible onset times satisfy $h \leq P - d$. That changes the event deliberately. Do not switch indexing conventions between model fitting and deployment. A stopped run with insufficient follow-up is censored or excluded according to a declared rule, rather than automatically labeled as no plateau.

5 An exact contraction equation trail

Suppose a supplied analytical reference has loss $L_{t+h} = L_t \rho^h$, with $0 < \rho < 1$. Its one-step improvement at offset h is

$$d_h = L_t(1 - \rho)\rho^h. \tag{5.1}$$

For an absolute-improvement plateau criterion $d_h \leq \varepsilon$, the first eligible nonnegative integer is

$$h_* = \max\left\{0, \left\lceil \frac{\log[\varepsilon/(L_t(1-\rho))]}{\log \rho} \right\rceil \right\}. \tag{5.2}$$

Proof. Divide by the positive initial improvement. Taking logarithms yields $h \log \rho \leq \log[\varepsilon/(L_t(1 - \rho))]$. Since $\log \rho < 0$, division reverses the inequality. Choose the smallest nonnegative integer satisfying it. □

At $\rho = 0$ or 1, calculate the sequence directly instead of dividing by $\log \rho$. This example supplies an exact law for explanation; the data-driven predictor must estimate behavior from observations and cannot treat a fitted ρ as certain.

Worked example 5.1 (A finite remaining-progress calculation)**.** Take $L_t = 1$, $\rho = 1/2$ and $\varepsilon = 1/16$. Improvement is $(1/2)^{h+1}$ and reaches the nonstrict boundary at h=3. If the criterion used strict improvement $< \varepsilon$, the first h would be 4. Boundary conventions change the answer.

Even exact contraction eventually produces small absolute improvements. Plateau morphology can therefore be a feature of healthy convergence, not just stagnation. An application interested in wasted compute needs a useful-improvement or utility target, not the morphology label alone.

6 Noise, slope and low-variability requirements

For a linear loss model $L_j = a + bj + \epsilon_j$ with independent equal-variance observation errors and fixed sample times, ordinary least-squares slope variance is $\sigma^2 / \sum_j (j - \bar{j})^2$. Correlation,

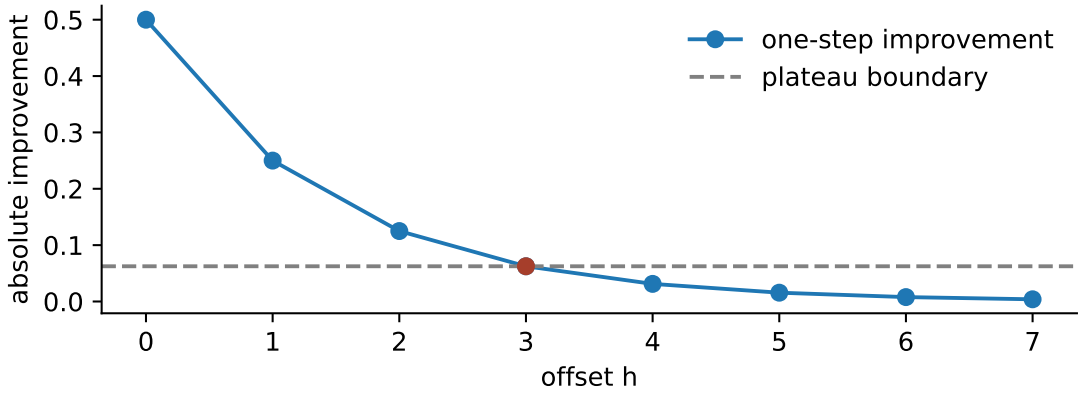


Figure 1: For loss contraction ratio 1/2, one-step improvement reaches the nonstrict boundary 1/16 at offset 3. This is a constructed limiting model, not an estimated countdown.

heteroscedasticity and overlapping windows change that expression. A small estimated slope with large uncertainty is not established flatness.

If future increments are independently $N(\mu, \sigma^2)$, the probability that d consecutive increments all satisfy $|\delta| \leq \varepsilon$ is

$$\left[\Phi\left(\frac{\varepsilon - \mu}{\sigma}\right) - \Phi\left(\frac{-\varepsilon - \mu}{\sigma}\right)\right]^d. \tag{6.1}$$

This product follows from independence. It is not valid unchanged for increments obtained by differencing shared noisy loss values, since adjacent increments then share an error term. A joint model or a directly fitted plateau-event predictor avoids an unjustified independence shortcut.

The morphology's slope and variability requirements address different aspects. Low variance can accompany a substantial drift; a near-zero slope can accompany large oscillation. Preserve both, and any declared persistence requirement, rather than substituting a single scalar "flatness" score without showing equivalence.

7 A zero gradient is not a proof of a good plateau

Proposition 7.1 (Stationarity does not identify an optimum)**.** *A zero gradient can occur at a local maximum, saddle or minimum. Therefore a gradient-only plateau criterion does not establish optimization success.*

Proof. For $L(\theta) = \theta^4 - \theta^2$, $L'(0) = 0$ but $L''(0) = -2$, so zero is a local maximum. For $L(x, y) = x^2 - y^2$, the origin is a saddle with zero gradient. For $L(\theta) = \theta^2$, it is a minimum. The gradient value is identical at all three stationary points. □

Likewise, training loss and validation quality can diverge. If the desired output is a chance of useful validation improvement, define $\mathbb{P}(L_t^{\text{val}} - L_{t+P}^{\text{val}} > \delta \mid \mathcal{F}_t)$ separately. A plateau probability cannot be relabeled as that probability without an additional model.

8 Recognition gates and selective risk

An applicability gate R_t may exclude states where the plateau forecasting model lacks support. It is distinct from the current morphology flag A_t. The reported probability can be restricted to $\{A_t = 0, R_t = 1\}$ with explicit abstention elsewhere. If the gate is a function of $\mathcal{F}_t$, it adds no new information; it defines the reporting subset.

Let coverage be $c = \mathbb{P}(R = 1)$ and accepted/rejected risks be R_A, R_B for a fixed predictor. Then total risk is $cR_A + (1 - c)R_B$ by the law of total expectation. With an abstention action of cost a and prediction loss ℓ, selective decision risk is

$$\mathbb{E}[R\ell + (1 - R)a]. \tag{8.1}$$

An ideal decision accepts where its conditional prediction risk is at most a. A fitted gate does not know that oracle quantity and requires qualification. Accuracy on a small accepted subset must be reported with its coverage, not extrapolated to rejected states.

9 Calibration may change on the accepted subset

A model can be calibrated marginally and not calibrated after selection. Suppose an observed group G has two equally common values with event probabilities 0.8 and 0.2. The constant score 0.5 is marginally calibrated. A gate accepting only the first group leaves mean outcome 0.8 at the same score 0.5, so accepted-subset calibration fails.

This example uses a legitimate observed gate, not future labels. It shows why a reliability statement must identify its population. If the model already supplies the true conditional probability given all gating information, calibration is preserved under that information; a weaker marginal statement does not supply it.

10 WPM construction and an application worksheet

Fix the plateau morphology and initially at-risk population. Build target-specific W,m predictors of future morphology or arrival as declared. Use only matured labels. A short-window pocket may represent recent flattening, while a long-window pocket captures stable trend; retain their flow, position and existence without assuming either is universally better.

Fit a direct event model or a joint predictive distribution for future slope and variability. If B, S are modeled jointly, plateau probability is $\mathbb{P}(|B| \le \varepsilon_b, S \le \varepsilon_s \mid \mathcal{F}_t)$. Multiplying marginal probabilities assumes the relevant conditional independence. A mixture of eligible same-target event forecasts is another defined model.

For a P=20 query using a future morphology window ending at 20, return probability, current morphology status, applicability gate, time units and uncertainty interpretation. If confirmation extends beyond 20, disclose the later maturity. If training is already plateaued, either report a separate persistence query or abstain from the entry query; do not claim a new transition forecast.

11 Telemetry comparison and scope

Scoped evidence statement. The archived compact comparison excluded already-plateaued states and reported stronger location-plus-scale forecasts than its specified telemetry comparator

for future plateau morphology [5]. Earlier recognition-gated plateau notes had limited accepted support, so these populations should not be conflated. A blanket closeness claim would obscure the target-specific advantage in that compact comparison; closeness to telemetry is supported for selected other AI metrics and contexts, not every plateau population. The classifier learned from observations, not from the optimizer's true future equations.

The mathematical method is portable, but event thresholds, observation precision, optimizer policies and definitions of useful improvement must be specified for each application. These forecasts do not establish a stopping action's causal benefit or the model's final generalization quality.

12 Rules and solved questions

1. Separate current plateau recognition, future endpoint morphology, first arrival and useful-progress decisions.
2. State slope, variability, units, tolerance and persistence requirements.
3. Include confirmation time in label maturity and handle censored runs explicitly.
4. Report gate coverage and accepted-population reliability together.
5. Preserve a separate action model when recommending stopping or schedule changes.

Exercise 12.1. Does a constant high-loss trajectory satisfy a low-slope, low-variance morphology?

Solution. It can. Morphology alone does not certify a good optimum.

Exercise 12.2. A query allows onset by step 20 followed by 5 confirming increments. When is every negative label mature?

Solution. At step 25 under this indexing convention, plus any reporting delay. An alternative within-horizon definition would restrict allowed onset times.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This study shares the AI prediction collection DOI 10.5281/zenodo.23784369.

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